SE – Lab 7

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5H

Before fixing the codes:

```
(.venv) C:\Users\srira\Documents>python mobile_data_plan.py
==== Running parse_plan ====
Enter plan:data in GB (e.g. standard:5): standard:5
Plan and Data: ('Standard', 5.0)

==== Running usage_average ====
Enter daily usage in MB as CSV (e.g. 300,450,800): 300
Average Usage: 300

==== Running payable_bill ====
Enter bill|discount (e.g. 399|50): 399|50
Payable Bill: 420.82

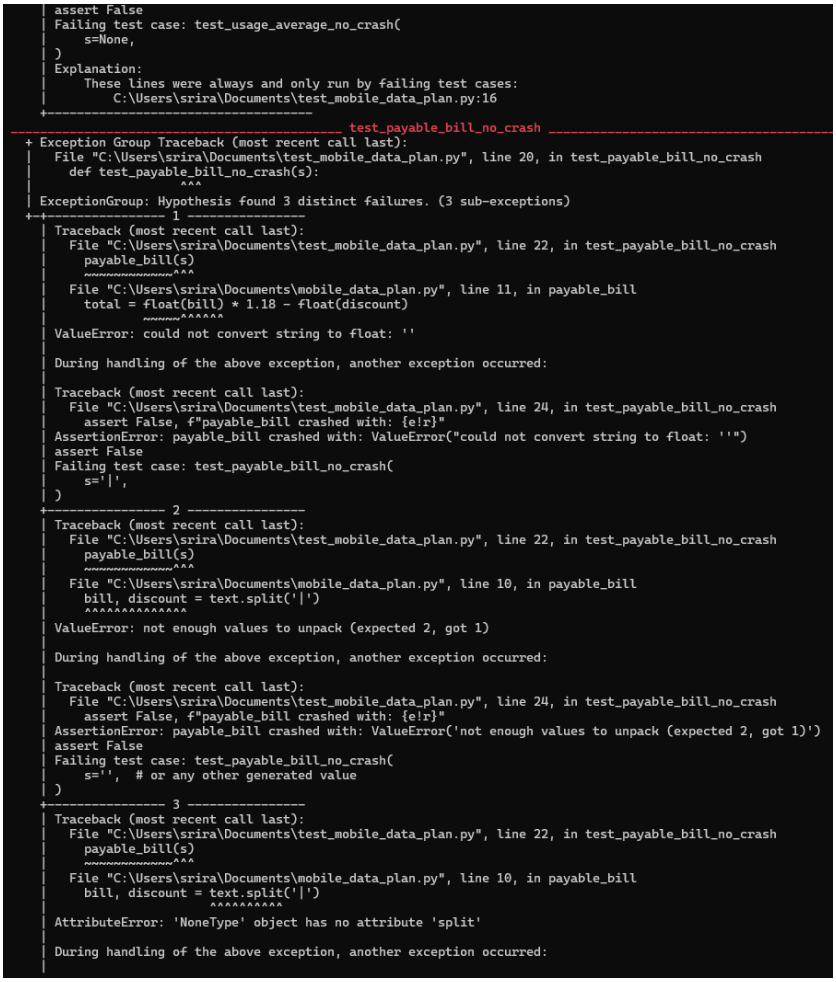
(.venv) C:\Users\srira\Documents>python mobile_data_plan.py
==== Running parse_plan ====
Enter plan:data in GB (e.g. standard:5): standard
Traceback (most recent call last):
  File "C:\Users\srira\Documents\mobile_data_plan.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\Users\srira\Documents\mobile_data_plan.py", line 17, in main
    result = parse_plan(user_input)
  File "C:\Users\srira\Documents\mobile_data_plan.py", line 2, in parse_plan
    plan, gb = text.split(':')
    ^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

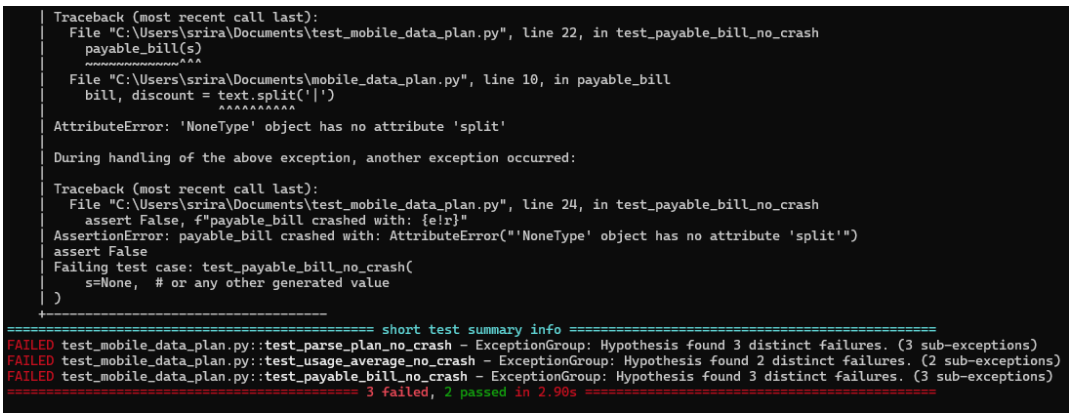
```
(.venv) C:\Users\srira\Documents>pytest test_mobile_data_plan.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\srira\Documents\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\srira\Documents
plugins: hypothesis-6.170.15
collected 5 items

test_mobile_data_plan.py::test_parse_plan_no_crash FAILED                                                [ 20%]
test_mobile_data_plan.py::test_usage_average_no_crash FAILED                                             [ 40%]
test_mobile_data_plan.py::test_payable_bill_no_crash FAILED                                              [ 60%]
test_mobile_data_plan.py::test_usage_average_valid_input PASSED                                          [ 80%]
test_mobile_data_plan.py::test_payable_bill_never_negative PASSED                                        [100%]

================================================== FAILURES ==================================================
_____________________________________________ test_parse_plan_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 6, in test_parse_plan_no_crash
  |     def test_parse_plan_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 8, in test_parse_plan_no_crash
    |     parse_plan(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\srira\Documents\mobile_data_plan.py", line 3, in parse_plan
    |     return plan.strip().title(), float(gb)
    |                                  ~~~~~^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 10, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with: {e!r}"
    | AssertionError: parse_plan crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 8, in test_parse_plan_no_crash
    |     parse_plan(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\srira\Documents\mobile_data_plan.py", line 2, in parse_plan
    |     plan, gb = text.split(':')
    |     ^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 10, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with: {e!r}"
    | AssertionError: parse_plan crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 8, in test_parse_plan_no_crash
```

```
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 8, in test_parse_plan_no_crash
    |     parse_plan(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\srira\Documents\mobile_data_plan.py", line 2, in parse_plan
    |     plan, gb = text.split(':')
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 10, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with: {e!r}"
    | AssertionError: parse_plan crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________ test_usage_average_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 13, in test_usage_average_no_crash
  |     def test_usage_average_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 15, in test_usage_average_no_crash
    |     usage_average(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "C:\Users\srira\Documents\mobile_data_plan.py", line 6, in usage_average
    |     values = [int(r) for r in usage_csv.split(',')]
    |                  ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 17, in test_usage_average_no_crash
    |     assert False, f"usage_average crashed with: {e!r}"
    | AssertionError: usage_average crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_usage_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\srira\Documents\test_mobile_data_plan.py:16
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 15, in test_usage_average_no_crash
    |     usage_average(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "C:\Users\srira\Documents\mobile_data_plan.py", line 6, in usage_average
    |     values = [int(r) for r in usage_csv.split(',')]
    |                              ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 17, in test_usage_average_no_crash
    |     assert False, f"usage_average crashed with: {e!r}"
    | AssertionError: usage_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
```


| assert False
| Failing test case: test_usage_average_no_crash(
| s=None,
|)
| Explanation:
| These lines were always and only run by failing test cases:
| C:\Users\srira\Documents\test_mobile_data_plan.py:16
+------------------------------------
_____ test_payable_bill_no_crash _____
+ Exception Group Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 20, in test_payable_bill_no_crash
| def test_payable_bill_no_crash(s):
| ^^^
| ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------- 1 ----------
| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 22, in test_payable_bill_no_crash
| payable_bill(s)
| ~~~~~~~~~~~~^^^
| File "C:\Users\srira\Documents\mobile_data_plan.py", line 11, in payable_bill
| total = float(bill) * 1.18 - float(discount)
| ~~~~~^^^^^^
| ValueError: could not convert string to float: ''
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 24, in test_payable_bill_no_crash
| assert False, f"payable_bill crashed with: {e!r}"
| AssertionError: payable_bill crashed with: ValueError("could not convert string to float: ''")
| assert False
| Failing test case: test_payable_bill_no_crash(
| s='|',
|)
+---------- 2 ----------
| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 22, in test_payable_bill_no_crash
| payable_bill(s)
| ~~~~~~~~~~~~^^^
| File "C:\Users\srira\Documents\mobile_data_plan.py", line 10, in payable_bill
| bill, discount = text.split('|')
| ^^^^^^^^^^^^^^
| ValueError: not enough values to unpack (expected 2, got 1)
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 24, in test_payable_bill_no_crash
| assert False, f"payable_bill crashed with: {e!r}"
| AssertionError: payable_bill crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
| assert False
| Failing test case: test_payable_bill_no_crash(
| s='', # or any other generated value
|)
+---------- 3 ----------
| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 22, in test_payable_bill_no_crash
| payable_bill(s)
| ~~~~~~~~~~~~^^^
| File "C:\Users\srira\Documents\mobile_data_plan.py", line 10, in payable_bill
| bill, discount = text.split('|')
| ^^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'split'
|
| During handling of the above exception, another exception occurred:
|


| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 22, in test_payable_bill_no_crash
| payable_bill(s)
| ~~~~~~~~~~~~^^^
| File "C:\Users\srira\Documents\mobile_data_plan.py", line 10, in payable_bill
| bill, discount = text.split('|')
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| AttributeError: 'NoneType' object has no attribute 'split'
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| During handling of the above exception, another exception occurred:
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| Traceback (most recent call last):
| File "C:\Users\srira\Documents\test_mobile_data_plan.py", line 24, in test_payable_bill_no_crash
| assert False, f"payable_bill crashed with: {e!r}"
| AssertionError: payable_bill crashed with: AttributeError("'NoneType' object has no attribute 'split'")
| assert False
| Failing test case: test_payable_bill_no_crash(
| s=None, # or any other generated value
|)
+------------------------------------
========== short test summary info ==========
FAILED test_mobile_data_plan.py::test_parse_plan_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_mobile_data_plan.py::test_usage_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_mobile_data_plan.py::test_payable_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========== 3 failed, 2 passed in 2.90s ==========

After fixing the codes:

```
(.venv) C:\Users\srira\Documents>python mobile_data_plan.py
==== Running parse_plan ====
Enter plan:data in GB (e.g. standard:5): standard
Plan and Data: Invalid input

==== Running usage_average ====
Enter daily usage in MB as CSV (e.g. 300,450,800): ,,
Average Usage: Invalid input

==== Running payable_bill ====
Enter bill|discount (e.g. 399|50): abc|54
Payable Bill: Invalid input
```

```
(.venv) C:\Users\srira\Documents>pytest test_mobile_data_plan.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\srira\Documents\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\srira\Documents
plugins: hypothesis-6.170.15
collected 5 items

test_mobile_data_plan.py::test_parse_plan_no_crash PASSED                                                                          [ 20%]
test_mobile_data_plan.py::test_usage_average_no_crash PASSED                                                                       [ 40%]
test_mobile_data_plan.py::test_payable_bill_no_crash PASSED                                                                        [ 60%]
test_mobile_data_plan.py::test_usage_average_valid_input PASSED                                                                    [ 80%]
test_mobile_data_plan.py::test_payable_bill_never_negative PASSED                                                                  [100%]

============================================================ 5 passed in 0.61s =============================================================
```